

Lab - Windows Task Manager

Objectives

In this lab, you will explore Task Manager and manage processes from within Task Manager.

Part 1: Working in the Processes tab

Part 2: Working in the Services tab

Part 3: Working in the Performance tab

Background / Scenario

The Task Manager is a system monitor program that provides information about the processes and programs running on a computer. It also allows the termination of processes and programs and modification of process priority.

Required Resources

- A Windows PC with internet access

Instructions

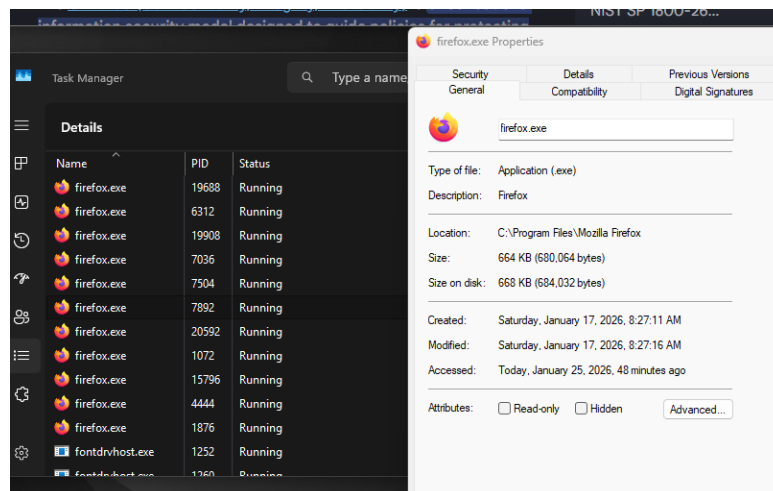
Part 1: Working in the Processes tab

- Open a command prompt and a web browser.

Microsoft Edge is used in this lab; however, any web browser will work. Just substitute your browser name whenever you see Microsoft Edge.

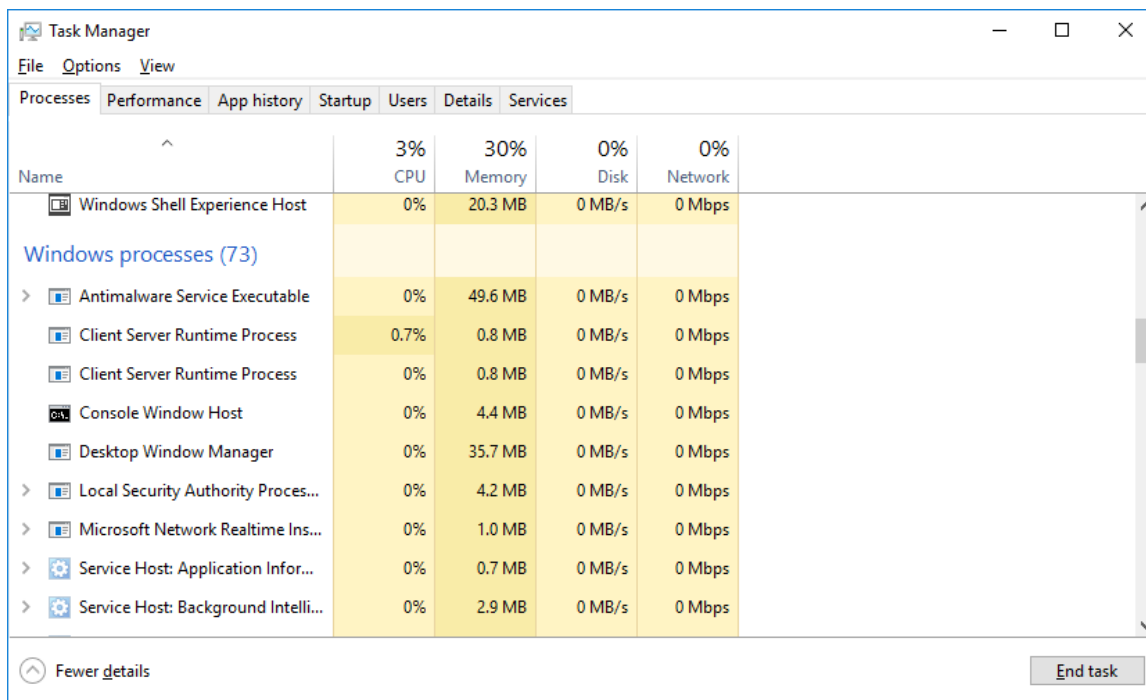
- Right-click the Task bar to open **Task Manager**. Another way to open the Task Manager is to press **Ctrl-Alt-Delete** to access the Windows Security screen and select **Task Manager**.
- Click **More details** to see all the processes that are listed in the Processes tab.
- Expand the Windows Command Processor heading.

What is listed under this heading?



- e. There are three categories of processes listed in the Processes tab: Apps, Background processes, and Windows processes.
- The Apps are the applications that you have opened, such as Microsoft Edge, Task Manager, and Windows Command Processor, as shown in the figure above. Other applications that are opened by the users, such as web browsers and email clients, will also be listed here.
 - The Background processes are executed in the background by applications that are currently open.
 - The Windows processes are not shown in the figure. Scroll down to view them on your Windows PC. Windows processes are Microsoft Windows services that run in the background.

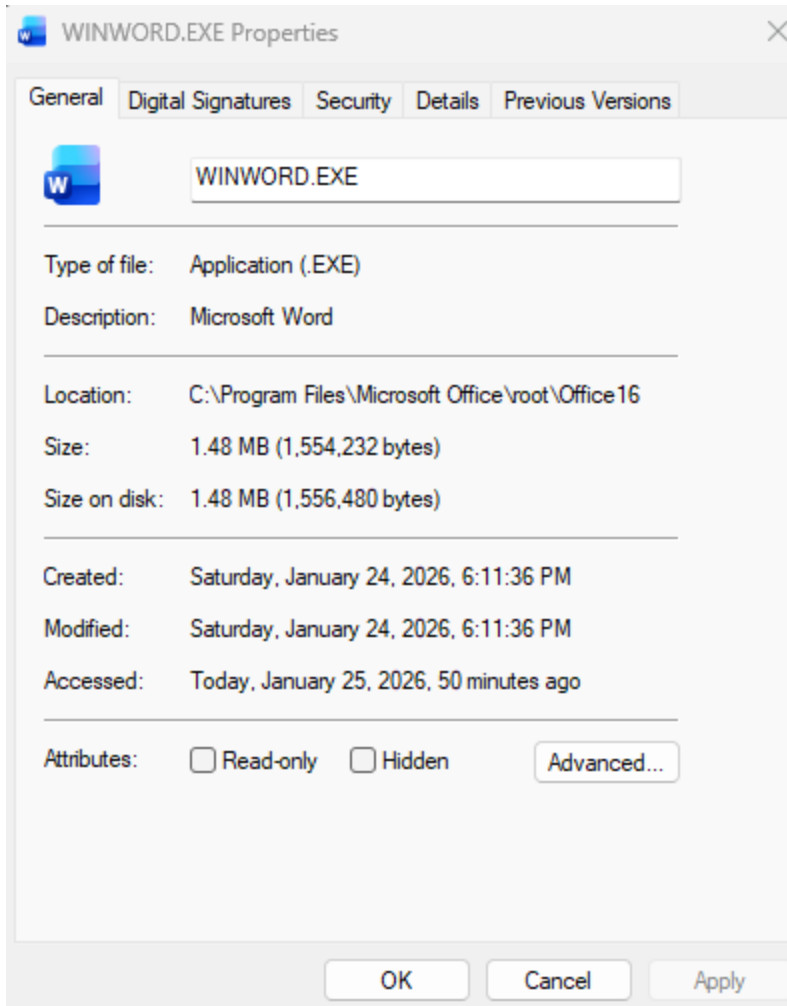
Some of the background processes or Windows processes may be associated with foreground processes. For example, if you open a command prompt window, the Console Window Host process will be started in the Windows process section, as shown below.



Task Manager				
File Options View				
Processes Performance App history Startup Users Details Services				
Name	3% CPU	30% Memory	0% Disk	0% Network
Windows Shell Experience Host	0%	20.3 MB	0 MB/s	0 Mbps
Windows processes (73)				
> Antimalware Service Executable	0%	49.6 MB	0 MB/s	0 Mbps
Client Server Runtime Process	0.7%	0.8 MB	0 MB/s	0 Mbps
Client Server Runtime Process	0%	0.8 MB	0 MB/s	0 Mbps
Console Window Host	0%	4.4 MB	0 MB/s	0 Mbps
Desktop Window Manager	0%	35.7 MB	0 MB/s	0 Mbps
> Local Security Authority Proces...	0%	4.2 MB	0 MB/s	0 Mbps
> Microsoft Network Realtime Ins...	0%	1.0 MB	0 MB/s	0 Mbps
> Service Host: Application Infor...	0%	0.7 MB	0 MB/s	0 Mbps
> Service Host: Background Intelli...	0%	2.9 MB	0 MB/s	0 Mbps
Fewer details				
End task				

- f. Right-click Console Window Host and select **Properties**.

What is the location of this filename and location of this process?



- g. Close the command prompt window.

What happens to Windows Command Processor and Console Window Host when the command prompt window is closed?

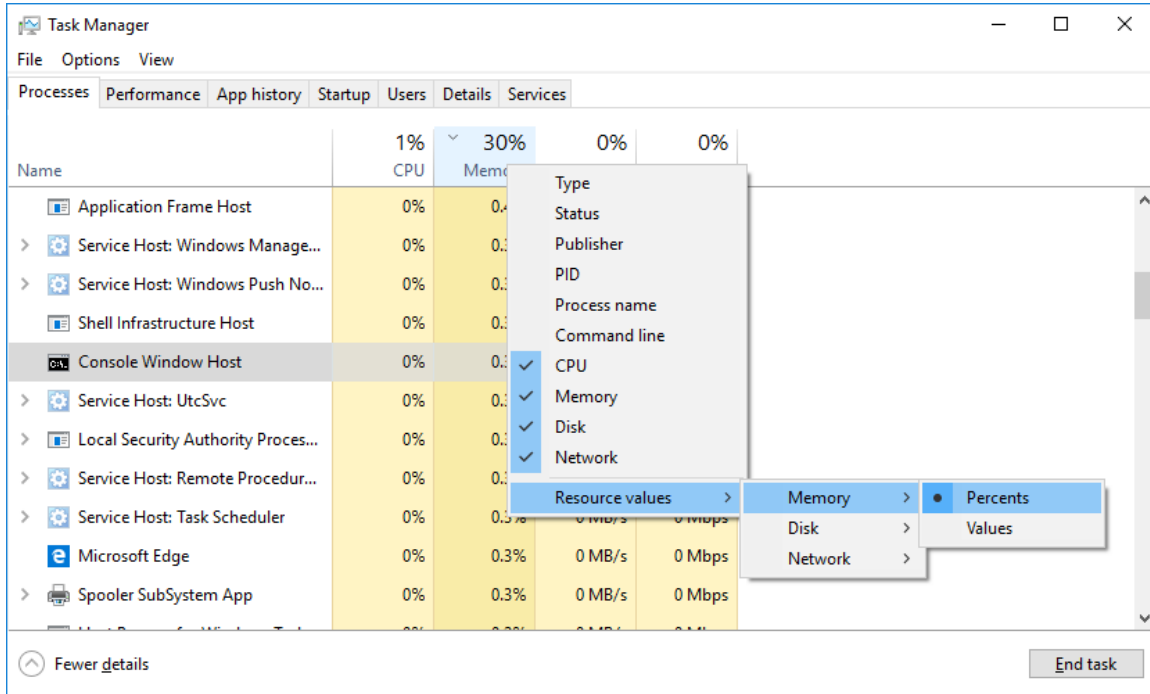
Answer: It deletes the task

- h. Click the Memory heading. Click the Memory heading a second time.

What effect does this have on the columns?

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- i. Right-click on the Memory heading, and then select **Resource values > Memory > Percents**.



What affect does this have on the Memory column?

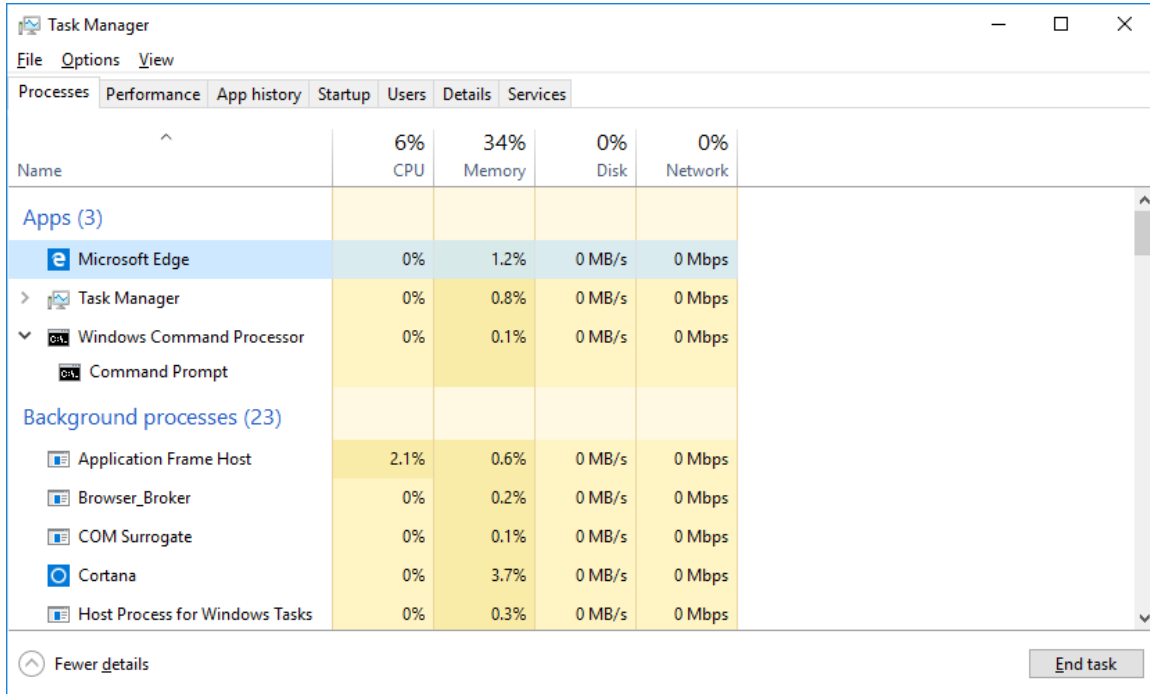
Processes		Run new task				End task	
Name	Status	0% CPU	44% Memory	6% Disk	1% Network		
> Media Player (2)		0%	5.2%	0 MB/s	0 Mbps		
> Windows Explorer		0%	3.5%	0 MB/s	0 Mbps		
> Antimalware Service Executable		0%	3.0%	0 MB/s	0 Mbps		

How could this be useful?

Answer: Know which program or application that eat or take up more space in RAM

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- j. In the Task Manager, click the **Name** heading.



- k. Double-click the Microsoft Edge.

What happens?

Processes						Run new task	End task
Name	Status	CPU	Memory	Disk	Network		
Apps (4)							
Media Player (2)		0%	4.8%	0 MB/s	0 Mbps		
Microsoft Word		0%	2.9%	0 MB/s	0.1 Mbps		
Task Manager		0%	1.6%	0 MB/s	0 Mbps		
Windows Explorer		7.1%	3.5%	0 MB/s	0 Mbps		

- l. Return to the Task Manager and right-click Microsoft Edge. Select **End task**.

What happens to the web browser windows?

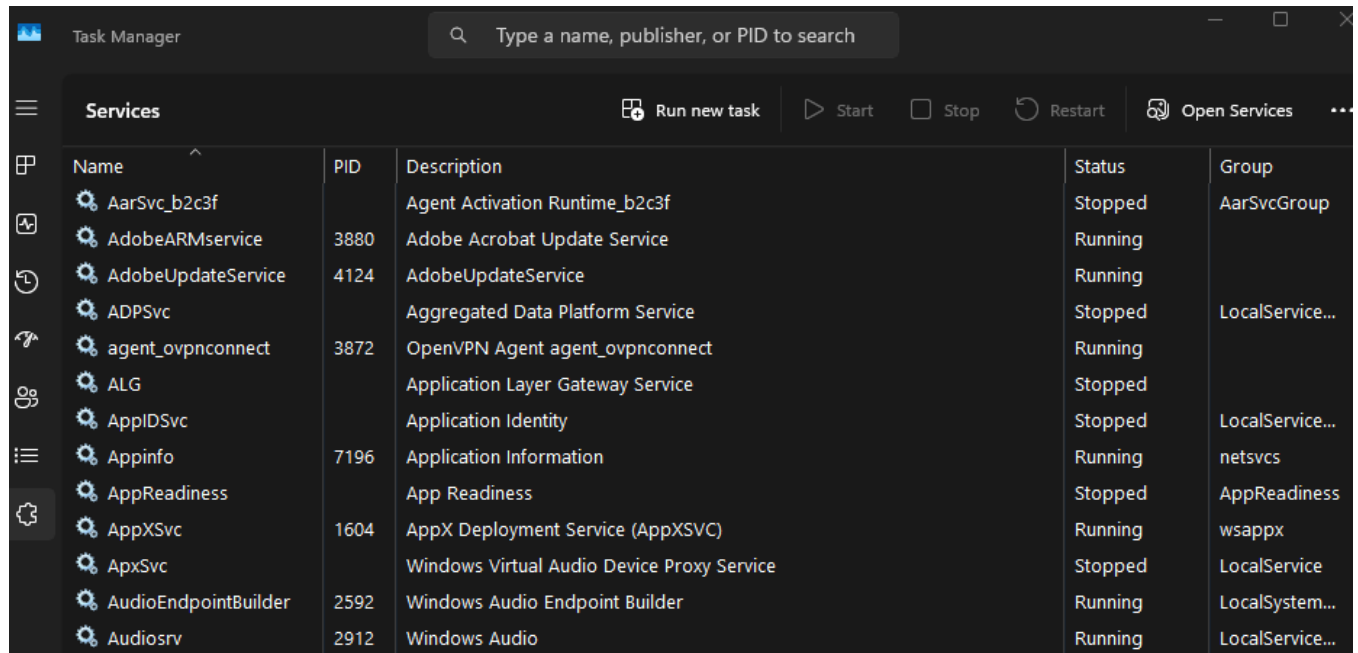
Answer: Close the application window

Part 2: Working in the Services tab

In the Task Manager window, click the **Services** tab. Use the scroll bar on the right side of the **Services** window to view all the services listed.

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What statuses are listed?

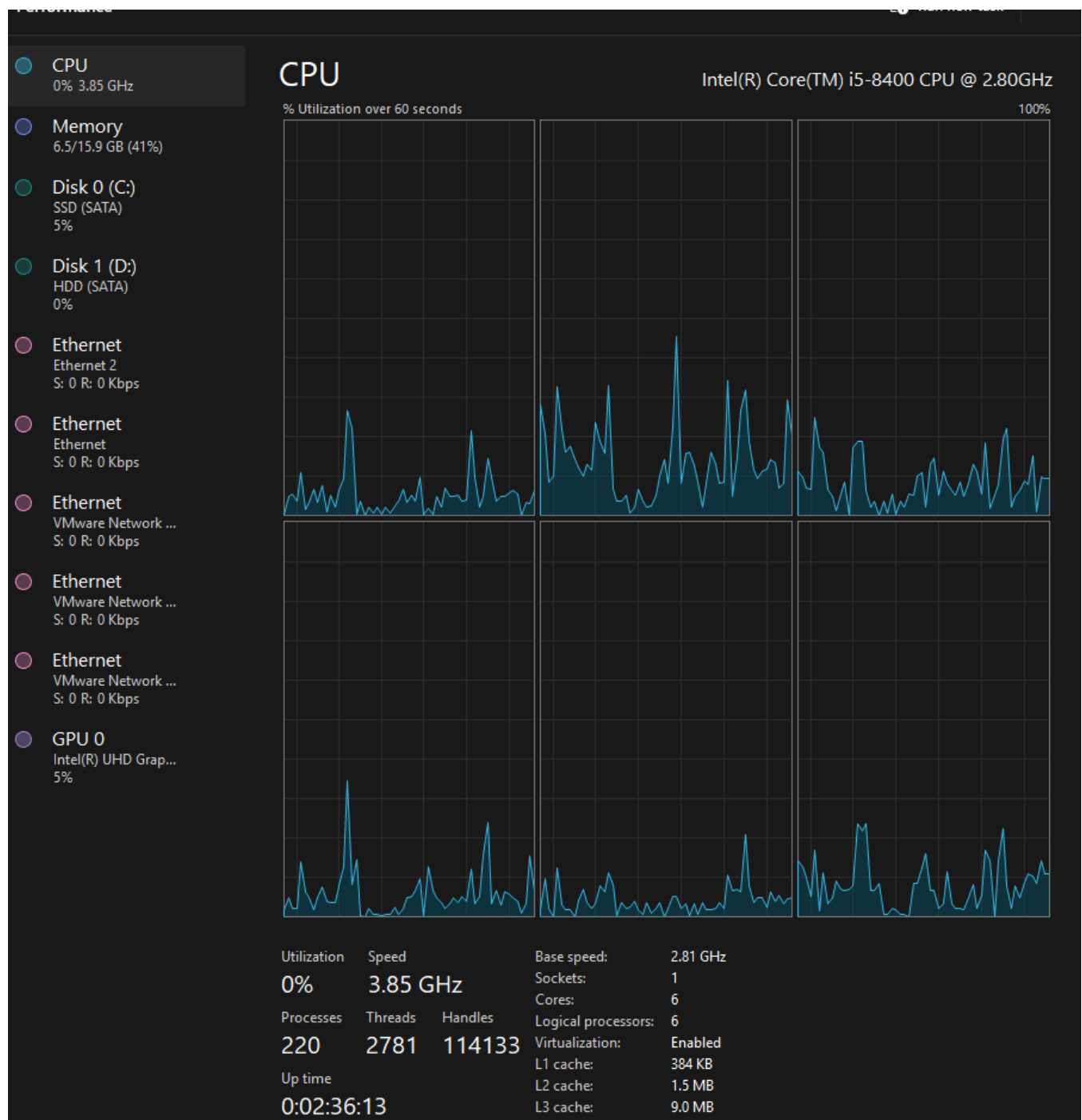


The screenshot shows the Windows Task Manager application with the 'Services' tab selected. The window title is 'Task Manager'. At the top, there is a search bar with the placeholder text 'Type a name, publisher, or PID to search'. Below the search bar, there are several action buttons: 'Run new task', 'Start', 'Stop', 'Restart', and 'Open Services'. The main area displays a list of services with columns for Name, PID, Description, Status, and Group. The services listed include AarSvc_b2c3f, AdobeARMSvc, AdobeUpdateService, ADPSvc, agent_ovpnconnect, ALG, AppIDSvc, Appinfo, AppReadiness, AppXSvc, ApxSvc, AudioEndpointBuilder, and Audiosrv. The status of each service is indicated in the 'Status' column, and the group it belongs to is in the 'Group' column.

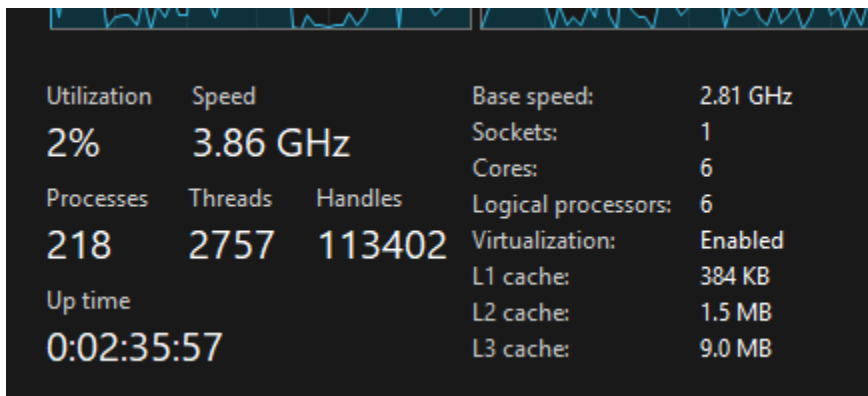
Name	PID	Description	Status	Group
AarSvc_b2c3f		Agent Activation Runtime_b2c3f	Stopped	AarSvcGroup
AdobeARMSvc	3880	Adobe Acrobat Update Service	Running	
AdobeUpdateService	4124	AdobeUpdateService	Running	
ADPSvc		Aggregated Data Platform Service	Stopped	LocalService...
agent_ovpnconnect	3872	OpenVPN Agent agent_ovpnconnect	Running	
ALG		Application Layer Gateway Service	Stopped	
AppIDSvc		Application Identity	Stopped	LocalService...
Appinfo	7196	Application Information	Running	netsvcs
AppReadiness		App Readiness	Stopped	AppReadiness
AppXSvc	1604	AppX Deployment Service (AppXSVC)	Running	wsappx
ApxSvc		Windows Virtual Audio Device Proxy Service	Stopped	LocalService
AudioEndpointBuilder	2592	Windows Audio Endpoint Builder	Running	LocalSystem...
Audiosrv	2912	Windows Audio	Running	LocalService...

Part 3: Working in the Performance tab

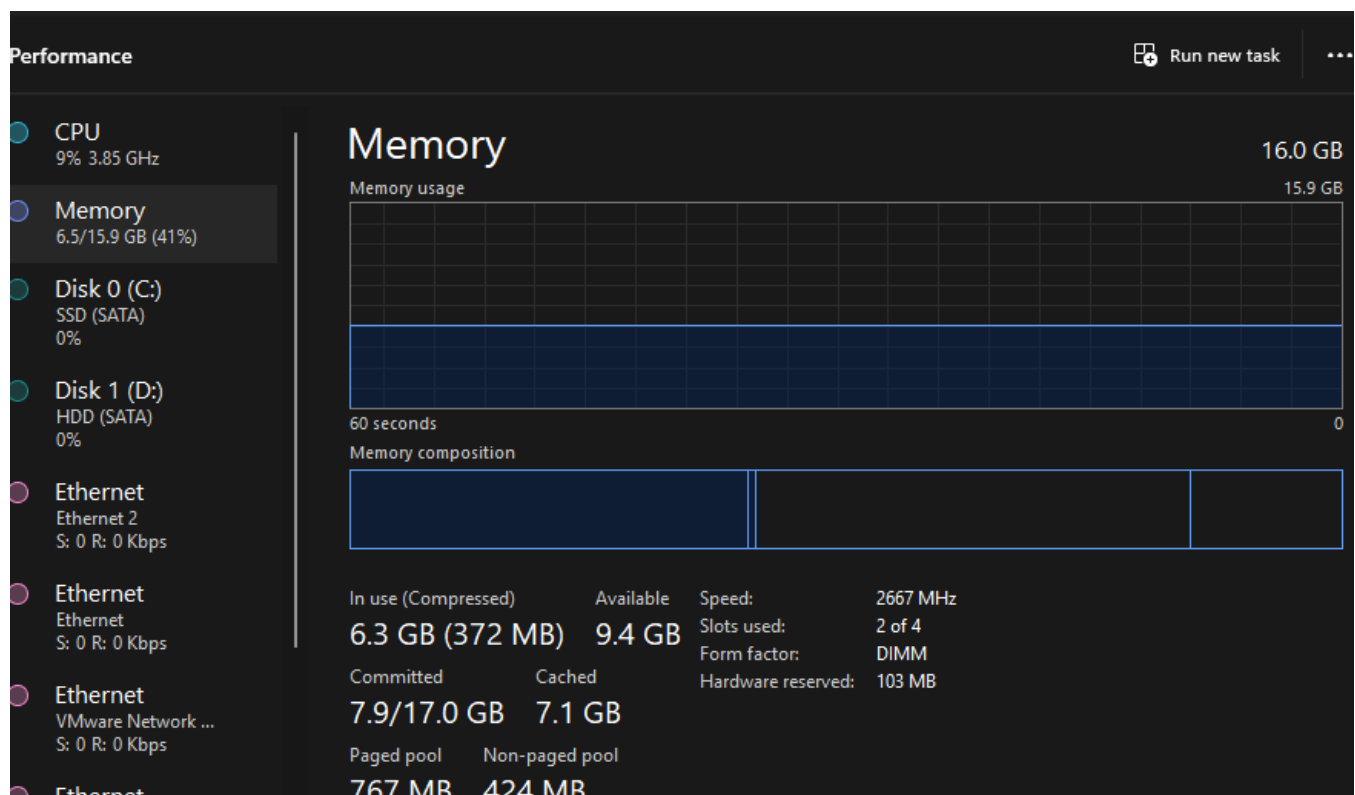
- In the Task Manager window, click the **Performance** tab.



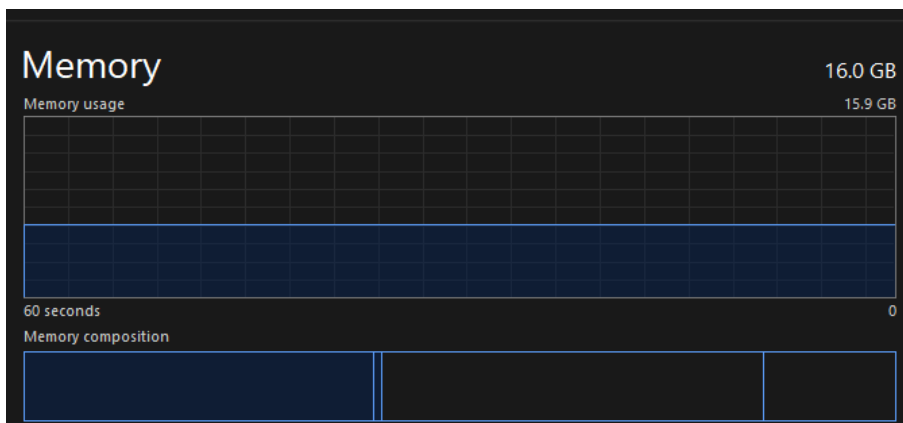
How many processes are running?



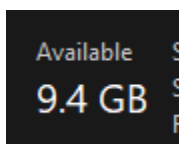
b. Click the **Memory** in the left panel of the **Performance** tab.



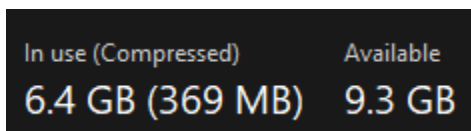
What is the total physical memory (MB)?



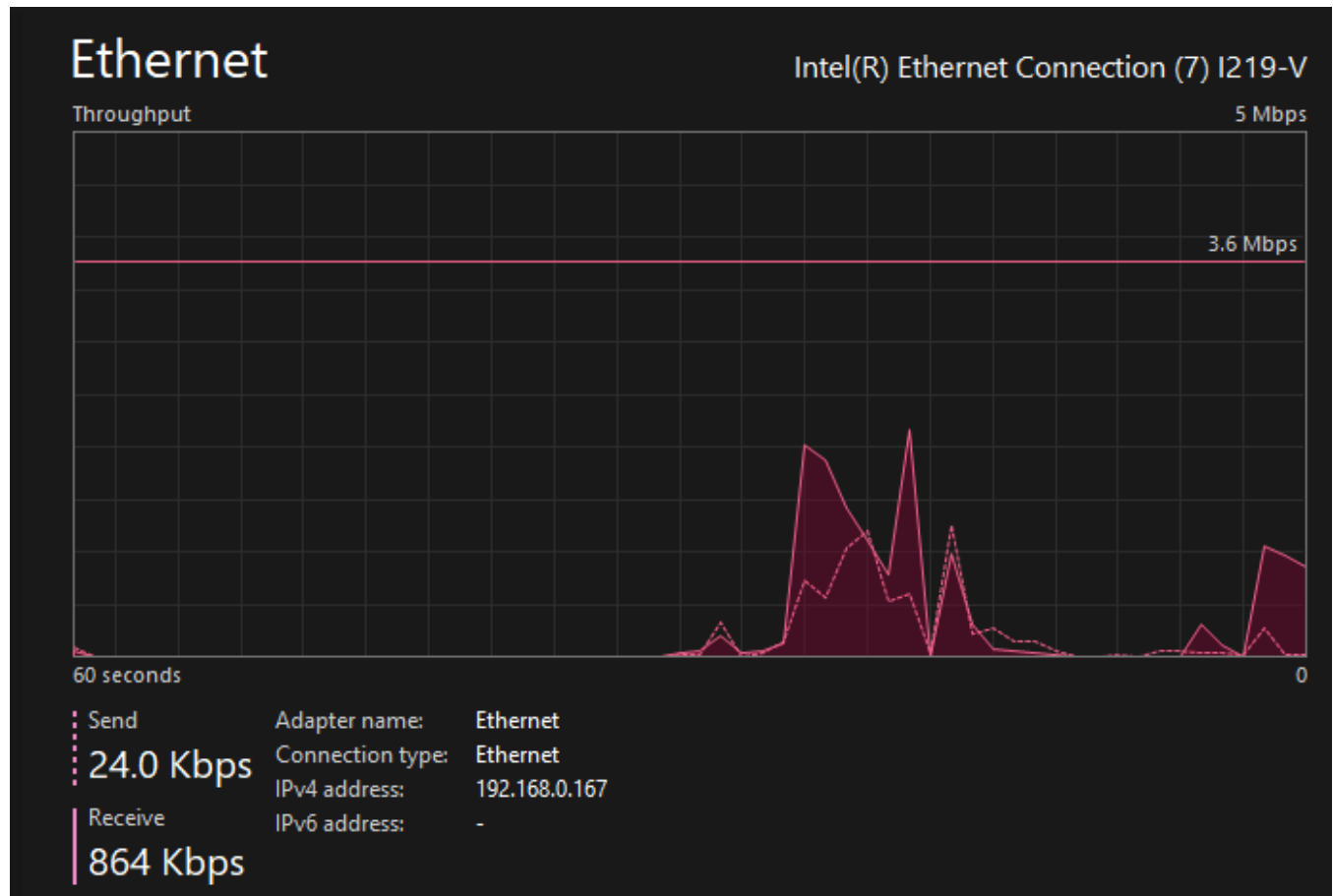
What is the available physical memory (MB)?



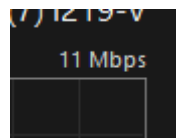
How much physical memory (MB) is being used by the computer?



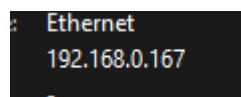
- c. Click the **Ethernet Chart** in the left panel of the **Performance** tab.



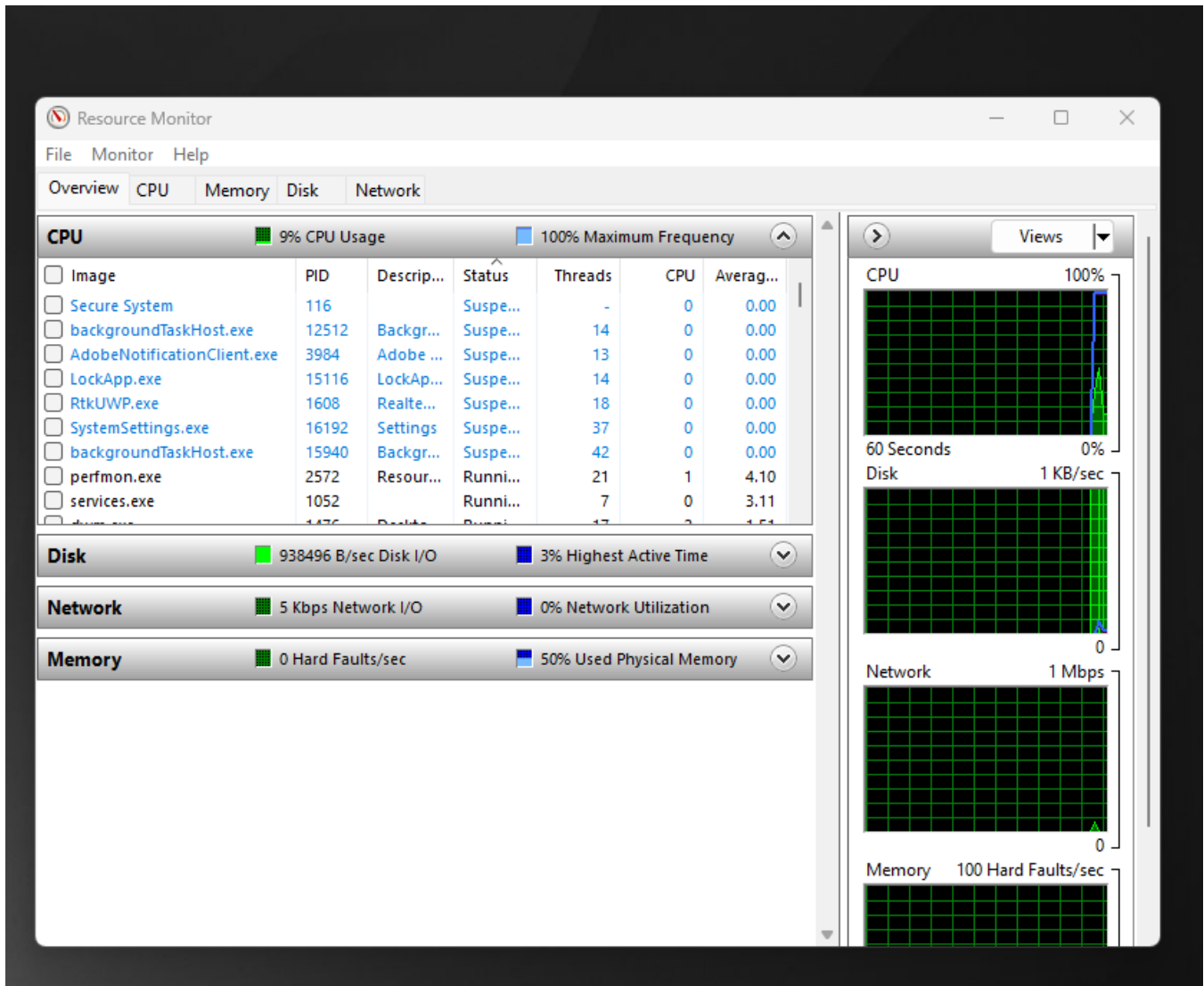
What is the link speed?



What is the IPv4 address of the PC?



- d. Click **Open Resource Monitor** to open the Resource Monitor utility from the Performance tab in Task Manager.



Reflection Question

Why is it important for an administrator to understand how to work within the Task Manager?

Answer: To know everything which running in your system and know which process take space a lot and which aren't both knowing what is active and not with the status of all of them.